

STAAD SPACE
START JOB INFORMATION
ENGINEER DATE April 2007
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN

JOINT COORDINATES

*MAIN HALL

1002	-52.000	3.200	-8.049;	1004	-52.000	5.600	-8.299;
1005	-52.000	7.419	-8.488;	1010	-52.000	11.585	-6.210;
1012	-52.000	11.756	-3.400;	1013	-52.000	11.885	0.000;
1014	-52.000	11.756	3.400;	1016	-52.000	11.585	6.210;
1021	-52.000	7.419	8.488;	1022	-52.000	5.600	8.299;
1024	-52.000	3.200	8.049;				
1101	-48.000	0.000	-7.715;	1102	-48.000	3.200	-8.051;
1103	-48.000	4.926	-8.233;	1104	-48.000	5.600	-8.304;
1105	-48.000	7.600	-8.514;	1106	-48.000	8.800	-8.640;
1107	-48.000	9.849	-8.543;	1108	-48.000	10.781	-8.055;
1109	-48.000	11.455	-7.243;	1110	-48.000	11.767	-6.237;
1111	-48.000	11.848	-4.900;	1112	-48.000	11.939	-3.400;
1113	-48.000	12.146	0.000;	1114	-48.000	11.939	3.400;
1115	-48.000	11.848	4.900;	1116	-48.000	11.767	6.237;
1117	-48.000	11.455	7.243;	1118	-48.000	10.781	8.055;
1119	-48.000	9.849	8.543;	1120	-48.000	8.800	8.640;
1121	-48.000	7.600	8.514;	1122	-48.000	5.600	8.304;
1123	-48.000	4.926	8.233;	1124	-48.000	3.200	8.051;
1125	-48.000	0.000	7.715;				
1202	-45.600	3.200	-8.051;	1204	-45.600	5.600	-8.304;
1205	-45.600	7.781	-8.540;	1210	-45.600	11.949	-6.264;
1212	-45.600	12.122	-3.400;	1213	-45.600	12.947	0.000;
1214	-45.600	12.122	3.400;	1216	-45.600	11.949	6.264;
1221	-45.600	7.781	8.540;	1222	-45.600	5.600	8.304;
1223	-45.600	4.926	8.237;	1224	-45.600	3.200	8.051;
1225	-45.600	0.000	7.715;				
1301	-40.000	0.000	-7.715;	1302	-40.000	3.200	-8.051;
1303	-40.000	5.033	-8.244;	1304	-40.000	5.600	-8.304;
1305	-40.000	7.859	-8.541;	1306	-40.000	9.059	-8.667;
1307	-40.000	10.108	-8.571;	1308	-40.000	11.040	-8.080;
1309	-40.000	11.714	-7.270;	1310	-40.000	12.027	-6.264;
1311	-40.000	12.110	-4.900;	1312	-40.000	12.201	3.400;
1313	-40.000	12.407	0.000;	1314	-40.000	12.201	3.400;
1315	-40.000	12.110	4.900;	1316	-40.000	12.027	6.264;
1317	-40.000	11.714	7.270;	1318	-40.000	11.040	8.080;
1319	-40.000	10.108	8.571;	1320	-40.000	9.059	8.667;
1321	-40.000	7.859	8.541;	1322	-40.000	5.600	8.304;
1323	-40.000	5.033	8.244;	1324	-40.000	3.200	8.051;
1325	-40.000	0.000	7.715;				
1401	-32.000	0.000	-7.715;	1402	-32.000	3.200	-8.051;
1403	-32.000	5.151	-8.257;	1404	-32.000	5.600	-8.304;
1405	-32.000	8.071	-8.564;	1406	-32.000	9.271	-8.690;
1407	-32.000	10.320	-8.593;	1408	-32.000	11.253	-8.103;
1409	-32.000	11.926	-7.293;	1410	-32.000	12.239	-6.287;
1411	-32.000	12.323	-4.900;	1412	-32.000	12.414	-3.400;
1413	-32.000	12.621	0.000;	1414	-32.000	12.414	3.400;
1415	-32.000	12.323	4.900;	1416	-32.000	12.239	6.287;
1417	-32.000	11.926	7.293;	1418	-32.000	11.253	8.103;
1419	-32.000	10.320	8.593;	1420	-32.000	9.271	8.690;
1421	-32.000	8.071	8.564;	1422	-32.000	5.600	8.293;
1423	-32.000	5.151	8.257;	1424	-32.000	3.200	8.051;
1425	-32.000	0.000	7.715;				
1501	-24.000	0.000	-7.715;	1502	-24.000	3.200	-8.051;
1503	-24.000	5.242	-8.266;	1504	-24.000	5.500	-8.304;
1505	-24.000	8.236	-8.581;	1506	-24.000	9.436	-8.707;
1507	-24.000	10.485	-8.610;	1508	-24.000	11.418	-8.112;
1509	-24.000	12.091	-7.310;	1510	-24.000	12.404	-6.304;
1511	-24.000	12.489	-4.900;	1512	-24.000	12.580	-3.400;
1513	-24.000	12.787	0.000;	1514	-24.000	12.580	3.400;
1515	-24.000	12.489	4.900;	1516	-24.000	12.404	6.304;

Diesel Engine Workshop3.std

1517	-24.000	12.091	7.310;	1518	-24.000	11.418	8.112;
1519	-24.000	10.485	8.610;	1520	-24.000	9.436	8.707;
1521	-24.000	8.236	8.581;	1522	-24.000	5.600	8.304;
1523	-24.000	5.242	8.266;	1524	-24.000	3.200	8.051;
1525	-24.000	0.000	7.715;				
1601	-16.000	0.000	-7.715;	1602	-16.000	3.200	-8.051;
1603	-16.000	5.308	-8.273;	1604	-16.000	5.600	-8.304;
1605	-16.000	8.354	-8.593;	1606	-16.000	9.554	-8.719;
1607	-16.000	10.603	-8.623;	1608	-16.000	11.536	-8.132;
1609	-16.000	12.209	-7.322;	1610	-16.000	12.522	-6.316;
1611	-16.000	12.608	-4.900;	1612	-16.000	12.699	-3.400;
1613	-16.000	12.905	0.000;	1614	-16.000	12.699	3.400;
1615	-16.000	12.608	4.900;	1616	-16.000	12.522	6.316;
1617	-16.000	12.209	7.322;	1618	-16.000	11.536	8.132;
1619	-16.000	10.603	8.623;	1620	-16.000	9.554	8.719;
1621	-16.000	8.354	8.593;	1622	-16.000	5.600	8.304;
1623	-16.000	5.308	8.273;	1624	-16.000	3.200	8.051;
1625	-16.000	0.000	7.715;				
1701	-8.000	0.000	-7.715;	1702	-8.000	3.200	-8.051;
1703	-8.000	5.347	-8.277;	1704	-8.000	5.600	-8.304;
1705	-8.000	8.425	-8.601;	1706	-8.000	9.625	-8.727;
1707	-8.000	10.674	-8.630;	1708	-8.000	11.606	-8.140;
1709	-8.000	12.280	-7.330;	1710	-8.000	12.592	-6.323;
1711	-8.000	12.678	-4.900;	1712	-8.000	12.770	-3.400;
1713	-8.000	12.976	0.000;	1714	-8.000	12.770	3.400;
1715	-8.000	12.678	4.900;	1716	-8.000	12.592	6.323;
1717	-8.000	12.280	7.330;	1718	-8.000	11.606	8.140;
1719	-8.000	10.674	8.630;	1720	-8.000	9.625	8.727;
1721	-8.000	8.425	8.601;	1722	-8.000	5.600	8.304;
1723	-8.000	5.347	8.277;	1724	-8.000	3.200	8.051;
1725	-8.000	0.000	7.715;				
1801	0.000	0.000	-7.715;	1802	0.000	3.200	-8.051;
1803	0.000	5.360	-8.278;	1804	0.000	5.600	-8.304;
1805	0.000	8.449	-8.603;	1806	0.000	9.649	-8.729;
1807	0.000	10.698	-8.633;	1808	0.000	11.630	-8.142;
1809	0.000	12.300	-7.332;	1810	0.000	12.616	-6.326;
1811	0.000	12.703	-4.900;	1812	0.000	12.794	-3.400;
1813	0.000	13.000	0.000;	1814	0.000	12.794	3.400;
1815	0.000	12.703	4.900;	1816	0.000	12.616	6.326;
1817	0.000	12.300	7.332;	1818	0.000	11.630	8.142;
1819	0.000	10.698	8.633;	1820	0.000	9.649	8.729;
1821	0.000	8.449	8.603;	1822	0.000	5.600	8.304;
1823	0.000	5.360	8.278;	1824	0.000	3.200	8.051;
1825	0.000	0.000	7.715;				
1901	8.000	0.000	-7.715;	1902	8.000	3.200	-8.051;
1903	8.000	5.347	-8.277;	1904	8.000	5.600	-8.304;
1905	8.000	8.425	-8.601;	1906	8.000	9.625	-8.727;
1907	8.000	10.674	-8.630;	1908	8.000	11.606	-8.140;
1909	8.000	12.280	-7.330;	1910	8.000	12.592	-6.323;
1911	8.000	12.678	-4.900;	1912	8.000	12.770	-3.400;
1913	8.000	12.976	0.000;	1914	8.000	12.770	3.400;
1915	8.000	12.678	4.900;	1916	8.000	12.592	6.323;
1917	8.000	12.280	7.330;	1918	8.000	11.606	8.140;
1919	8.000	10.674	8.630;	1920	8.000	9.625	8.727;
1921	8.000	8.425	8.601;	1922	8.000	5.600	8.304;
1923	8.000	5.347	8.277;	1924	8.000	3.200	8.051;
1925	8.000	0.000	7.715;				
2001	16.000	0.000	-7.715;	2002	16.000	3.200	-8.051;
2003	16.000	5.308	-8.273;	2004	16.000	5.600	-8.304;
2005	16.000	8.354	-8.593;	2006	16.000	9.554	-8.719;
2007	16.000	10.603	-8.623;	2008	16.000	11.536	-8.132;
2009	16.000	12.209	-7.322;	2010	16.000	12.522	-6.316;
2011	16.000	12.608	-4.900;	2012	16.000	12.699	-3.400;
2013	16.000	12.905	0.000;	2014	16.000	12.699	3.400;
2015	16.000	12.608	4.900;	2016	16.000	12.522	6.316;
2017	16.000	12.209	7.322;	2018	16.000	11.536	8.132;
2019	16.000	10.603	8.623;	2020	16.000	9.554	8.719;
2021	16.000	8.354	8.593;	2022	16.000	5.600	8.304;

2203	16.000	5.308	8.273;	2024	16.000	3.200	8.051;
2205	16.000	0.000	7.715;				
2101	24.000	0.000	-7.715;	2102	24.000	3.200	-8.051;
2103	24.000	5.242	-8.266;	2104	24.000	5.600	-8.304;
2105	24.000	8.236	-8.581;	2106	24.000	9.436	-8.707;
2107	24.000	10.485	-8.610;	2108	24.000	11.418	-8.112;
2109	24.000	12.091	-7.310;	2110	24.000	12.404	-6.304;
2111	24.000	12.489	-4.900;	2112	24.000	12.580	-4.300;
2113	24.000	12.787	0.000;	2114	24.000	12.580	3.400;
2115	24.000	12.489	4.900;	2116	24.000	12.404	6.304;
2117	24.000	12.091	7.310;	2118	24.000	11.418	8.112;
2119	24.000	10.485	8.610;	2120	24.000	9.436	8.707;
2121	24.000	8.236	8.581;	2122	24.000	5.600	8.304;
2123	24.000	5.242	8.266;	2124	24.000	3.200	8.051;
2125	24.000	0.000	7.715;				
2201	32.000	0.000	-7.715;	2202	32.000	3.200	-8.051;
2203	32.000	5.151	-8.257;	2204	32.000	5.600	-8.304;
2205	32.000	8.071	-8.584;	2206	32.000	9.271	-8.103;
2207	32.000	10.320	-8.593;	2208	32.000	11.253	-8.103;
2209	32.000	11.926	-7.293;	2210	32.000	12.239	-6.287;
2211	32.000	12.323	-4.900;	2212	32.000	12.414	-3.400;
2213	32.000	12.323	0.000;	2214	32.000	12.414	3.400;
2215	32.000	12.323	4.900;	2216	32.000	12.239	6.287;
2217	32.000	11.926	7.293;	2218	32.000	11.253	8.103;
2219	32.000	10.320	8.593;	2220	32.000	9.271	8.103;
2221	32.000	8.071	8.584;	2222	32.000	5.600	8.304;
2223	32.000	5.151	8.257;	2224	32.000	3.200	8.051;
2225	32.000	0.000	7.715;				
2301	40.000	0.000	-7.715;	2302	40.000	3.200	-8.051;
2303	40.000	5.033	-8.244;	2304	40.000	5.600	-8.304;
2305	40.000	7.859	-8.541;	2306	40.000	9.059	-8.667;
2307	40.000	10.108	-8.571;	2308	40.000	11.040	-8.080;
2309	40.000	11.714	-7.270;	2310	40.000	12.027	-6.264;
2311	40.000	12.110	-4.900;	2312	40.000	12.201	-3.400;
2313	40.000	12.407	0.000;	2314	40.000	12.201	3.400;
2315	40.000	12.110	4.900;	2316	40.000	12.027	6.264;
2317	40.000	11.714	7.270;	2318	40.000	11.040	8.080;
2319	40.000	10.108	8.571;	2320	40.000	9.059	8.667;
2321	40.000	7.859	8.541;	2322	40.000	5.600	8.304;
2323	40.000	5.033	8.244;	2324	40.000	3.200	8.051;
2325	40.000	0.000	7.715;				
2402	45.600	3.200	-8.051;	2404	45.600	5.600	-8.304;
2405	45.600	7.781	-8.540;	2410	45.600	11.949	-6.264;
2412	45.600	12.122	-3.400;	2413	45.600	12.407	0.000;
2414	45.600	12.122	3.400;	2416	45.600	11.949	6.264;
2421	45.600	7.781	8.540;	2422	45.600	5.600	8.304;
2423	45.600	4.926	8.237;	2424	45.600	3.200	8.051;
2425	45.600	0.000	7.715;				
2501	48.000	0.000	-7.715;	2502	48.000	3.200	-8.051;
2503	48.000	4.926	-8.233;	2504	48.000	5.600	-8.304;
2505	48.000	7.600	-8.514;	2506	48.000	8.800	-8.640;
2507	48.000	9.849	-8.543;	2508	48.000	10.781	-8.053;
2509	48.000	11.455	-7.243;	2510	48.000	11.767	-6.237;
2511	48.000	11.848	-4.900;	2512	48.000	11.939	-3.400;
2513	48.000	12.146	0.000;	2514	48.000	11.939	3.400;
2515	48.000	11.848	4.900;	2516	48.000	11.767	6.237;
2517	48.000	11.455	7.243;	2518	48.000	10.781	8.053;
2519	48.000	9.849	8.543;	2520	48.000	8.800	8.640;
2521	48.000	7.600	8.514;	2522	48.000	5.600	8.304;
2523	48.000	4.926	8.233;	2524	48.000	3.200	8.051;
2525	48.000	0.000	7.715;				
2602	52.000	3.200	-8.049;	2604	52.000	5.600	-8.299;
2605	52.000	7.419	-8.488;	2610	52.000	11.585	-6.210;
2612	52.000	11.756	-3.400;	2613	52.000	11.885	0.000;
2614	52.000	11.756	3.400;	2616	52.000	11.585	6.210;
2621	52.000	7.419	8.488;	2622	52.000	5.600	8.299;
2624	52.000	3.200	8.049;				

*OFFICE AREA							
1231	-45.600	4.926	8.600;	1232	-45.600	4.926	11.000;
1233	-45.600	4.926	13.318;	1234	-45.600	0.000	12.800;
1331	-40.000	5.033	8.600;	1332	-40.000	5.033	11.000;
1333	-40.000	5.033	13.329;	1334	-40.000	0.000	12.800;
1431	-32.000	5.151	8.600;	1432	-32.000	5.151	11.000;
1433	-32.000	5.151	13.341;	1434	-32.000	0.000	12.800;
1531	-24.000	5.242	8.600;	1532	-24.000	5.242	11.000;
1533	-24.000	5.242	13.351;	1534	-24.000	0.000	12.800;
1631	-16.000	5.308	8.600;	1632	-16.000	5.308	11.000;
1633	-16.000	5.308	13.358;	1634	-16.000	0.000	12.800;
1731	-8.000	5.347	8.600;	1732	-8.000	5.347	11.000;
1733	-8.000	5.347	13.362;	1734	-8.000	0.000	12.800;
1831	0.000	5.360	8.600;	1832	0.000	5.360	11.000;
1833	0.000	5.360	13.363;	1834	0.000	0.000	12.800;
1931	8.000	5.347	8.600;	1932	8.000	5.347	11.000;
1933	8.000	5.347	13.362;	1934	8.000	0.000	12.800;
2031	16.000	5.308	8.600;	2032	16.000	5.308	11.000;
2033	16.000	5.308	13.358;	2034	16.000	0.000	12.800;
2131	24.000	5.242	8.600;	2132	24.000	5.242	11.000;
2133	24.000	5.242	13.351;	2134	24.000	0.000	12.800;
2231	32.000	5.151	8.600;	2232	32.000	5.151	11.000;
2233	32.000	5.151	13.341;	2234	32.000	0.000	12.800;
2331	40.000	5.033	8.600;	2332	40.000	5.033	11.000;
2333	40.000	5.033	13.329;	2334	40.000	0.000	12.800;
2431	45.600	4.926	8.600;	2432	45.600	4.926	11.000;
2433	45.600	4.926	13.318;	2434	45.600	0.000	12.800;
*PERIMETER COLUMN							
3011	-48.000	0.000	-4.900;	3013	-48.000	0.000	0.000;
3015	-48.000	0.000	4.900;	3021	-48.000	3.200	-4.900;
3023	-48.000	3.200	0.000;	3025	-48.000	3.200	4.900;
3031	-48.000	7.600	-4.900;	3033	-48.000	7.600	0.000;
3035	-48.000	7.600	4.900;	3041	-48.000	8.800	-4.900;
3043	-48.000	8.800	0.000;	3045	-48.000	8.800	4.900;
3111	48.000	0.000	-4.900;	3113	48.000	0.000	0.000;
3115	48.000	0.000	4.900;	3121	48.000	3.200	-4.900;
3123	48.000	3.200	0.000;	3125	48.000	3.200	4.900;
3131	48.000	7.600	-4.900;	3133	48.000	7.600	0.000;
3135	48.000	7.600	4.900;	3141	48.000	8.800	-4.900;
3143	48.000	8.800	0.000;	3145	48.000	8.800	4.900;
MEMBER INCIDENCES							
*MAIN HALL MAIN FRAME							
1101	1101	1102	1124;	1222	1222	1223	1224;
1301	1301	1302	1324;	1401	1401	1402	1424;
1601	1601	1602	1624;	1701	1701	1702	1724;
1901	1901	1902	1924;	2001	2001	2002	2024;
2201	2201	2202	2224;	2301	2301	2302	2324;
2422	2422	2423	2424;	2501	2501	2502	2524;
*SECONDARY BEAM							
4001	1010	1110;	4002	1110	1210;	4003	1210
4005	1410	1510;	4006	1510	1610;	4007	1610
4009	1810	1910;	4010	1910	2010;	4011	2010
4013	2210	2310;	4014	2310	2410;	4015	2410
4101	1012	1112;	4102	1112	1212;	4103	1212
4105	1412	1512;	4106	1512	1612;	4107	1612
4109	1812	1912;	4110	1912	2012;	4111	2012
4113	2212	2312;	4114	2312	2412;	4115	2412
4201	1013	1113;	4202	1113	1213;	4203	1213
4205	1413	1513;	4206	1513	1613;	4207	1613
4209	1813	1913;	4210	1913	2013;	4211	2013
4213	2213	2313;	4214	2313	2413;	4215	2413
4301	1014	1114;	4302	1114	1214;	4303	1214
4305	1414	1514;	4306	1514	1614;	4307	1614
4309	1814	1914;	4310	1914	2014;	4311	2014
4313	2214	2314;	4314	2314	2414;	4315	2414

4401 1016 1116; 4402 1116 1216; 4403 1216 1316; 4404 1316 1416;
4405 1416 1516; 4406 1516 1616; 4407 1616 1716; 4408 1716 1816;
4409 1816 1916; 4410 1916 2016; 4411 2016 2116; 4412 2116 2216;
4413 2216 2316; 4414 2316 2416; 4415 2416 2516; 4416 2516 2616;

*SIDE BEAM
5001 1002 1102; 5002 1102 1202; 5003 1202 1302; 5004 1302 1402;
5005 1402 1502; 5006 1502 1602; 5007 1602 1702; 5008 1702 1802;
5009 1802 1902; 5010 1902 2002; 5011 2002 2102; 5012 2102 2202;
5013 2202 2302; 5014 2302 2402; 5015 2402 2502; 5016 2502 2602;

5101 1004 1104; 5102 1104 1204; 5103 1204 1304; 5104 1304 1404;
5105 1404 1504; 5106 1504 1604; 5107 1604 1704; 5108 1704 1804;
5109 1804 1904; 5110 1904 2004; 5111 2004 2104; 5112 2104 2204;
5113 2204 2304; 5114 2304 2404; 5115 2404 2504; 5116 2504 2604;

5201 1005 1105; 5202 1105 1205; 5203 1205 1305; 5204 1305 1405;
5205 1405 1505; 5206 1505 1605; 5207 1605 1705; 5208 1705 1805;
5209 1805 1905; 5210 1905 2005; 5211 2005 2105; 5212 2105 2205;
5213 2205 2305; 5214 2305 2405; 5215 2405 2505; 5216 2505 2605;

5301 1021 1121; 5302 1121 1221; 5303 1221 1321; 5304 1321 1421;
5305 1421 1521; 5306 1521 1621; 5307 1621 1721; 5308 1721 1821;
5309 1821 1921; 5310 1921 2021; 5311 2021 2121; 5312 2121 2221;
5313 2221 2321; 5314 2321 2421; 5315 2421 2521; 5316 2521 2621;

5401 1022 1122; 5402 1122 1222; 5403 1222 1322; 5404 1322 1422;
5405 1422 1522; 5406 1522 1622; 5407 1622 1722; 5408 1722 1822;
5409 1822 1922; 5410 1922 2022; 5411 2022 2122; 5412 2122 2222;
5413 2222 2322; 5414 2322 2422; 5415 2422 2522; 5416 2522 2622;

5501 1024 1124; 5502 1124 1224; 5503 1224 1324; 5504 1324 1424;
5505 1424 1524; 5506 1524 1624; 5507 1624 1724; 5508 1724 1824;
5509 1824 1924; 5510 1924 2024; 5511 2024 2124; 5512 2124 2224;
5513 2224 2324; 5514 2324 2424; 5515 2424 2524; 5516 2524 2624;

*OFFICE
1230 1223 1231; 1231 1231 1232 1233; 1330 1323 1331;
1230 1331 1332 1333; 1430 1423 1431; 1431 1431 1432 1433;
1530 1523 1531; 1531 1531 1532 1533; 1630 1623 1631;
1631 1631 1632 1633; 1730 1723 1731; 1731 1731 1732 1733;
1830 1823 1831; 1831 1831 1832 1833; 1930 1923 1931;
1931 1931 1932 1933; 2030 2023 2031; 2031 2031 2032 2033;
2130 2123 2131; 2131 2131 2132 2133; 2230 2223 2231;
2231 2231 2232 2233; 2330 2323 2331; 2331 2331 2332 2333;
2430 2423 2431; 2431 2431 2432 2433;

*SECONDARY BEAM

6001 1231 1331; 6002 1331 1431; 6003 1431 1531; 6004 1531 1631;
6005 1631 1731; 6006 1731 1831; 6007 1831 1931; 6008 1931 2031;
6009 2031 2131; 6010 2131 2231; 6011 2231 2331; 6012 2331 2431;

6101 1232 1332; 6102 1332 1432; 6103 1432 1532; 6104 1532 1632;
6105 1632 1732; 6106 1732 1832; 6107 1832 1932; 6108 1932 2032;
6109 2032 2132; 6110 2132 2232; 6111 2232 2332; 6112 2332 2432;

6201 1233 1333; 6202 1333 1433; 6203 1433 1533; 6204 1533 1633;
6205 1633 1733; 6206 1733 1833; 6207 1833 1933; 6208 1933 2033;
6209 2033 2133; 6210 2133 2233; 6211 2233 2333; 6212 2333 2433;

*PERIMETER COLUMN

110 3011 3021; 111 3021 3031; 112 3031 3041; 113 3041 1111;
120 3013 3023; 121 3023 3033; 122 3033 3043; 123 3043 1113;
130 3015 3025; 131 3025 3035; 132 3035 3045; 133 3045 1115;
210 3111 3121; 211 3121 3131; 212 3131 3141; 213 3141 2511;
220 3113 3123; 221 3123 3133; 222 3133 3143; 223 3143 2513;
230 3115 3125; 231 3125 3135; 232 3135 3145; 233 3145 2515;

*MAIN HALL - ROOF BRACING
7001 1005 1110; 7002 1105 1010; 7003 1012 1113; 7004 1112 1013;
7005 1013 1114; 7006 1113 1014; 7007 1016 1121; 7008 1116 1021;

7101 1105 1310; 7102 1305 1110; 7103 1110 1312; 7104 1310 1112;
7105 1112 1313; 7106 1312 1113; 7107 1113 1314; 7108 1313 1114;
7109 1114 1316; 7110 1314 1116; 7111 1116 1321; 7112 1316 1121;

7201 1805 1910; 7202 1905 1810; 7203 1810 1912; 7204 1910 1812;
7205 1812 1913; 7206 1912 1813; 7207 1813 1914; 7208 1913 1814;
7209 1814 1916; 7210 1914 1816; 7211 1816 1921; 7212 1916 1821;

7301 1905 2010; 7302 2005 1910; 7303 1910 2012; 7304 2010 1912;
7305 1912 2013; 7306 2012 1913; 7307 1913 2014; 7308 2013 1914;
7309 1914 2016; 7310 2014 1916; 7311 1916 2021; 7312 2016 1921;

7401 2305 2510; 7402 2505 2310; 7403 2310 2512; 7404 2510 2312;
7405 2312 2513; 7406 2512 2313; 7407 2313 2514; 7408 2513 2314;
7409 2314 2516; 7410 2514 2316; 7411 2316 2521; 7412 2516 2321;

7501 2505 2610; 7502 2605 2510; 7503 2512 2613; 7504 2612 2513;
7505 2513 2614; 7506 2613 2514; 7507 2516 2621; 7508 2616 2521;

*VERTICAL BRACING
7601 1101 1302; 7602 1301 1102; 7603 1102 1304; 7604 1302 1104;
7605 1104 1305; 7606 1304 1105; 7607 1121 1322; 7608 1321 1122;
7609 1122 1224; 7610 1222 1124; 7611 1124 1225; 7612 1224 1125;
7613 1222 1324; 7614 1322 1224; 7615 1224 1325; 7616 1324 1225;

7701 1801 1902; 7702 1901 1802; 7703 1802 1904; 7704 1902 1804;
7705 1804 1905; 7706 1904 1805; 7707 1821 1922; 7708 1921 1822;
7709 1822 1924; 7710 1922 1824; 7711 1824 1925; 7712 1924 1825;

7801 1901 2002; 7802 2001 1902; 7803 1902 2004; 7804 2002 1904;
7805 1904 2005; 7806 2004 1905; 7807 1921 2022; 7808 2021 1922;
7809 1922 2024; 7810 2022 1924; 7811 1924 2025; 7812 2024 1925;

7901 2301 2502; 7902 2501 2302; 7903 2302 2504; 7904 2502 2304;
7905 2304 2505; 7906 2504 2305; 7907 2321 2522; 7908 2521 2322;
7909 2322 2424; 7910 2422 2324; 7911 2324 2425; 7912 2424 2325;
7913 2422 2524; 7914 2522 2424; 7915 2424 2525; 7916 2524 2425;

*PERIMETER BRACING
8001 1101 3021; 8002 3011 1102; 8003 1102 3031; 8004 3021 1105;
8005 1105 3041; 8006 3031 1106;
8007 1124 3035; 8008 3025 1121; 8009 1121 3045; 8010 3035 1120;
8011 1102 3021; 8012 1105 3031; 8013 1106 3041;
8014 1120 3045; 8015 1121 3035; 8016 1124 3025;

8101 2501 3121; 8102 3111 2502; 8103 2502 3131; 8104 3121 2505;
8105 2505 3141; 8106 3131 2506;
8107 2524 3135; 8108 3125 2521; 8109 2521 3145; 8110 3135 2520;
8111 2502 3121; 8112 2505 3131; 8113 2506 3141;
8114 2520 3145; 8115 2521 3135; 8116 2524 3125;

*OFFICE BRACING
9001 1231 1332; 9002 1331 1232; 9003 1232 1333; 9004 1332 1233;
9005 1233 1334; 9006 1333 1234;

9101 1831 1932; 9102 1931 1832; 9103 1832 1933; 9104 1932 1833;
9105 1833 1934; 9106 1933 1834;

9201 1931 2032; 9202 2031 1932; 9203 1932 2033; 9204 2032 1933;

9301 2331 2432; 9302 2431 2332; 9303 2332 2433; 9304 2432 2333;
9305 2333 2434; 9306 2433 2334;

9401 1223 1234; 9402 1225 1233; 9403 2423 2434; 9404 2425 2433;

DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03

END DEFINE MATERIAL

MEMBER PROPERTY BRITISH

*MAIN HALL - MAIN FRAME

1101 1301 1401 1501 1601 1701 1801 1901 2001 2101 2201 2301 2501 -
TAPERED 0.7 0.010 0.8 0.25 0.015 0.25 0.015
1102 1302 1402 1502 1602 1702 1802 1902 2002 2102 2202 2302 2502 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1103 1303 1403 1503 1603 1703 1803 1903 2003 2103 2203 2303 2503 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1104 1304 1404 1504 1604 1704 1804 1904 2004 2104 2204 2304 2504 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1105 1305 1405 1505 1605 1705 1805 1905 2005 2105 2205 2305 2505 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1106 1306 1406 1506 1606 1706 1806 1906 2006 2106 2206 2306 2506 -
TAPERED 0.8 0.010 0.75 0.25 0.015 0.25 0.015
1107 1307 1407 1507 1607 1707 1807 1907 2007 2107 2207 2307 2507 -
TAPERED 0.75 0.010 0.7 0.25 0.015 0.25 0.015
1108 1308 1408 1508 1608 1708 1808 1908 2008 2108 2208 2308 2508 -
TAPERED 0.7 0.010 0.65 0.25 0.015 0.25 0.015
1109 1309 1409 1509 1609 1709 1809 1909 2009 2109 2209 2309 2509 -
TAPERED 0.65 0.010 0.6 0.25 0.015 0.25 0.015
1110 TO 1115 -
1510 TO 1515 -
1610 TO 1615 1710 TO 1715 1810 TO 1815 1910 TO 1915 -
2010 TO 2015 2110 TO 2115 -
2510 TO 2515 TAPERED 0.6 0.008 0.6 0.2 0.012 0.2 0.012
*

1310 1410 2210 2310 -
TAPERED 0.6 0.008 0.7 0.2 0.012 0.2 0.012

*

1311 1411 2211 2311 -
TAPERED 0.7 0.008 0.8 0.2 0.012 0.2 0.012

*

1312 1412 2212 2312 -
TAPERED 0.8 0.008 1.0 0.2 0.012 0.2 0.012

*

1313 1413 2213 2313 -
TAPERED 1.0 0.008 0.8 0.2 0.012 0.2 0.012

*

1314 1414 2214 2314 -
TAPERED 0.8 0.008 0.7 0.2 0.012 0.2 0.012

*

1315 1415 2215 2315 -
TAPERED 0.7 0.008 0.6 0.2 0.012 0.2 0.012

*

1116 1316 1416 1516 1616 1716 1816 1916 2016 2116 2216 2316 2516 -
TAPERED 0.6 0.010 0.65 0.25 0.015 0.25 0.015
1117 1317 1417 1517 1617 1717 1817 1917 2017 2117 2217 2317 2517 -
TAPERED 0.65 0.010 0.7 0.25 0.015 0.25 0.015
1118 1318 1418 1518 1618 1718 1818 1918 2018 2118 2218 2318 2518 -
TAPERED 0.7 0.010 0.75 0.25 0.015 0.25 0.015
1119 1319 1419 1519 1619 1719 1819 1919 2019 2119 2219 2319 2519 -
TAPERED 0.75 0.010 0.8 0.25 0.015 0.25 0.015
1120 1320 1420 1520 1620 1720 1820 1920 2020 2120 2220 2320 2520 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1121 1321 1421 1521 1621 1721 1821 1921 2021 2121 2221 2321 2521 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1122 1322 1422 1522 1622 1722 1822 1922 2022 2122 2222 2322 2522 -
TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1123 1323 1423 1523 1623 1723 1823 1923 2023 2123 2223 2323 2523 -

TAPERED 0.8 0.010 0.8 0.25 0.015 0.25 0.015
1124 1324 1424 1524 1624 1724 1824 1924 2024 2124 2224 2324 2524 -
TAPERED 0.8 0.010 0.7 0.25 0.015 0.25 0.015
1222 TO 1224 -
2422 TO 2424 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
*MAIN HALL SECONDARY BEAMS
4001 4016 4101 4116 4201 4216 4301 4316 4401 4416 -
TAPERED 0.3 0.006 0.3 0.15 0.01 0.15 0.01
4002 TO 4015 4102 TO 4115 4202 TO 4215 4302 TO 4315 -
4402 TO 4415 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
*SIDE BEAM
5001 TO 5016 5101 TO 5116 5201 TO 5216 5301 TO 5316 5401 TO 5416 -
5501 TO 5516 TAPERED 0.3 0.010 0.3 0.25 0.15 0.25 0.15
*OFFICE MAIN ROOF BEAM
1230 TO 1232 1330 TO 1332 1430 TO 1432 1530 TO 1532 -
1630 TO 1632 1730 TO 1732 1830 TO 1832 1930 TO 1932 -
2030 TO 2032 2130 TO 2132 2230 TO 2232 2330 TO 2332 -
2430 TO 2432 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
*OFFICE SECONDARY BEAM
6001 TO 6012 6101 TO 6112 -
6201 TO 6212 TAPERED 0.4 0.006 0.4 0.15 0.01 0.15 0.01
*PERIMETER COLUMN
110 TO 113 130 TO 133 210 TO 213 230 TO 233 -
TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
120 TO 123 220 TO 223 TAPERED 0.3 0.010 0.3 0.3 0.015 0.3 0.015
*BRACING
7001 TO 7008 7101 TO 7112 7201 TO 7212 7301 TO 7312 -
7401 TO 7412 7501 TO 7508 7601 TO 7616 7701 TO 7712 -
7801 TO 7812 7901 TO 7916 8001 TO 8016 8101 TO 8116 -
9001 TO 9006 9101 TO 9106 9201 TO 9204 9301 TO 9306 TABLE ST UA90X90X6
9401 TO 9404 TABLE ST UA90X90X6
MEMBER RELEASES
4004 TO 4013 4104 TO 4113 4204 TO 4213 4304 TO 4313 -
4404 TO 4413 BOTH MX MY MZ
4003 4103 4203 4303 4403 END MX MY MZ
4014 4114 4214 4314 4414 START MX MY MZ
6001 TO 6012 6101 TO 6112 6201 TO 6212 BOTH MX MY MZ
1230 1330 1430 1530 1630 1730 1830 1930 2030 2130 -
2230 2330 2430 START MX MY MZ
MEMBER CABLE
7001 TO 7008 7101 TO 7112 7201 TO 7212 7301 TO 7312 -
7401 TO 7412 7501 TO 7508 7601 TO 7616 7701 TO 7712 -
7801 TO 7812 7901 TO 7916 8001 TO 8016 8101 TO 8116 -
9001 TO 9006 9101 TO 9106 9201 TO 9204 9301 TO 9306 -
9401 TO 9404 TENSION 5
CONSTANTS
MATERIAL STEEL ALL
BETA 90 MEMB 5001 TO 5016 5101 TO 5116 5201 TO 5216 5301 TO 5316 -
5401 TO 5416 5501 TO 5516
SUPPORTS
1101 1125 1225 1301 1325 1401 1425 1501 1525 1601 1625 -
1701 1725 1801 1825 1901 1925 2001 2025 2101 2125 2201 2225 -
2301 2325 2425 2501 2525 PINNED
1234 1334 1434 1534 1634 1734 1834 1934 -
2034 2134 2234 2334 2434 PINNED
3011 3013 3015 3111 3113 3115 PINNED
LOAD 1 SELFWEIGHT
SELFWEIGHT Y -1
LOAD 2 SDL

MEMBER LOAD
*MAIN HALL
4001 TO 4016 4401 TO 4416 UNI GY -4.62
4101 TO 4116 4301 TO 4316 UNI GY -3.16
4201 TO 4216 UNI GY -3.41
5001 TO 5016 5501 TO 5516 UNI GY -2.19
5101 TO 5116 5401 TO 5416 UNI GY -1.27
5201 TO 5216 5301 TO 5316 UNI GY -1.82
*OFFICE
6001 TO 6012 UNI GY -1.53
6101 TO 6112 UNI GY -2.38
6201 TO 6212 UNI GY -3.88

*AHU PLATFORM LOAD
1305 1405 2205 2305 CON GY -20.3 0.8
1312 1412 2212 2312 CON GY -20.3 0.275
1313 1413 2213 2313 CON GY -20.3 3.125
1320 1420 2220 2320 CON GY -20.3 0.4

LOAD 3 LL
MEMBER LOAD
*MAIN HALL
4001 TO 4016 4401 TO 4416 UNI GY -2.77
4101 TO 4116 4301 TO 4316 UNI GY -1.90
4201 TO 4216 UNI GY -2.05
*OFFICE
6001 TO 6012 UNI GY -0.92
6101 TO 6112 UNI GY -1.43
6201 TO 6212 UNI GY -0.71
*AIRCON LOAD
4204 TO 4213 CON GY -5
4203 CON GY -5 1.6
4214 CON GY -5 4.0
6101 TO 6112 CON GY -5

*AHU PLATFORM LOAD
1305 1405 2205 2305 CON GY -26.3 0.8
1312 1412 2212 2312 CON GY -26.3 0.275
1313 1413 2213 2313 CON GY -26.3 3.125
1320 1420 2220 2320 CON GY -26.3 0.4

*CRANE LOAD
JOINT LOAD
1804 FY -236.8
1804 FZ 35.5

LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
MEMBER LOAD
*MAIN HALL
4001 TO 4016 UNI GY 3.88
4101 TO 4116 UNI GY 2.65
4201 TO 4216 UNI GY 1.84
4401 TO 4416 UNI GY 0.76
4301 TO 4316 UNI GY 1.11
5001 TO 5016 UNI GZ 6.18
5101 TO 5116 UNI GZ 3.57
5201 TO 5216 UNI GZ 5.13
5301 TO 5316 UNI GZ 0.91
5401 TO 5416 UNI GZ 0.63
5501 TO 5516 UNI GZ 1.10
*OFFICE
6001 TO 6012 UNI GY 1.32
6101 TO 6112 UNI GY 1.62
6201 TO 6212 UNI GY 0.59
1233 2433 UNI GZ 1.00
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 2.00

LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
MEMBER LOAD
4001 TO 4016 UNI GY 6.75
4101 TO 4116 UNI GY 4.61
4201 TO 4216 UNI GY 3.34

4401 TO 4416 UNI GY 1.58
4301 TO 4316 UNI GY 2.31
5001 TO 5016 UNI GZ 3.46
5101 TO 5116 UNI GZ 2.00
5201 TO 5216 UNI GZ 2.88
5301 TO 5316 UNI GZ 3.17
5401 TO 5416 UNI GZ 2.20
5501 TO 5516 UNI GZ 3.81
*OFFICE
6001 TO 6012 UNI GY 2.26
6101 TO 6112 UNI GY 2.36
6201 TO 6212 UNI GY 0.59
1233 2433 UNI GZ 3.48
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 6.96

LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
MEMBER LOAD
4001 TO 4016 UNI GY -2.86
4101 TO 4116 UNI GY -1.96
4201 TO 4216 UNI GY -2.11
4401 TO 4416 UNI GY -1.96
4301 TO 4316 UNI GY -2.86
5001 TO 5016 UNI GZ -2.72
5101 TO 5116 UNI GZ -1.57
5201 TO 5216 UNI GZ -2.26
5301 TO 5316 UNI GZ 2.26
5401 TO 5416 UNI GZ 1.57
5501 TO 5516 UNI GZ 2.72
*OFFICE
6001 TO 6012 UNI GY -0.95
6101 TO 6112 UNI GY -1.48
6201 TO 6212 UNI GY -0.73
1233 2433 UNI GZ 2.48
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 4.96

LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
MEMBER LOAD
4001 TO 4016 UNI GY 5.17
4101 TO 4116 UNI GY 3.54
4201 TO 4216 UNI GY 3.82
4401 TO 4416 UNI GY 3.54
4301 TO 4316 UNI GY 5.17
5001 TO 5016 UNI GZ -5.43
5101 TO 5116 UNI GZ -3.14
5201 TO 5216 UNI GZ -4.51
5301 TO 5316 UNI GZ 4.51
5401 TO 5416 UNI GZ 3.14
5501 TO 5516 UNI GZ 5.43
*OFFICE
6001 TO 6012 UNI GY 1.71
6101 TO 6112 UNI GY 2.67
6201 TO 6212 UNI GY 1.32
1233 2433 UNI GZ 4.96
1333 1433 1533 1633 1733 1833 1933 2033 2133 -
2233 2333 UNI GZ 9.92

LOAD 21 EQ1 (EARTHQUAKE LOAD)
JOINT LOAD
1109 1115 1309 1315 1409 1415 1509 1515 1609 1615 1709 1715 -
1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
2509 2515 FX 28.5
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 FX 19.1

LOAD 22 EQ1 (EARTHQUAKE LOAD)
JOINT LOAD
1109 1115 1309 1315 1409 1415 1509 1515 1609 1615 1709 1715 -
1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
2509 2515 FZ 28.5
1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
2433 FZ 19.1

LOAD COMB 1001 1.4DL + 1.6LL
1 1.4 2 1.4 3 1.6

LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
1 1.2 2 1.2 3 1.2 11 1.2
LOAD COMB 1111 1.4DL + 1.4WL1
1 1.4 2 1.4 11 1.4
LOAD COMB 1211 1.0DL + 1.4WL1
1 1.0 2 1.0 11 1.4
LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
1 1.2 2 1.2 3 1.2 12 1.2
LOAD COMB 1112 1.4DL + 1.4WL2
1 1.4 2 1.4 12 1.4
LOAD COMB 1212 1.0DL + 1.4WL2
1 1.0 2 1.0 12 1.4
LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
1 1.2 2 1.2 3 1.2 13 1.2
LOAD COMB 1113 1.4DL + 1.4WL3
1 1.4 2 1.4 13 1.4
LOAD COMB 1213 1.0DL + 1.4WL3
1 1.0 2 1.0 13 1.4
LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
1 1.2 2 1.2 3 1.2 14 1.2
LOAD COMB 1114 1.4DL + 1.4WL4
1 1.4 2 1.4 14 1.4
LOAD COMB 1214 1.0DL + 1.4WL4
1 1.0 2 1.0 14 1.4

LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
1 1.2 2 1.2 3 0.5 21 1.0
LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
1 1.2 2 1.2 3 0.5 22 1.0
LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
1 1.2 2 1.2 3 0.5 21 -1.0
LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
1 1.2 2 1.2 3 0.5 22 -1.0
LOAD COMB 1221 0.9DL + 1.0EQ1
1 0.9 2 0.9 21 1.0
LOAD COMB 1222 0.9DL + 1.0EQ2
1 0.9 2 0.9 22 1.0
LOAD COMB 1321 0.9DL - 1.0EQ1
1 0.9 2 0.9 21 -1.0
LOAD COMB 1322 0.9DL - 1.0EQ2
1 0.9 2 0.9 22 -1.0
*SLS
LOAD COMB 2001 1.0DL + 1.0LL
1 1.0 2 1.0 3 1.0
LOAD COMB 2002 1.0DL + 0.5LL
1 1.0 2 1.0 3 0.5
LOAD COMB 2011 DL + LL + WL1
1 1.0 2 1.0 3 1.0 11 1.0
LOAD COMB 2012 DL + LL + WL2
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 2013 DL + LL + WL3
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 2014 DL + LL + WL4
1 1.0 2 1.0 3 1.0 14 1.0

PERFORM ANALYSIS
LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
1211 TO 1214 1221 1222 1321 1322
PRINT SUPPORT REACTION ALL
PRINT FORCE ENVELOPE NSECTION 5 ALL
PARAMETER 1
CODE BS5950
SBLT 1.0
PY 345000 ALL
CHECK CODE ALL
PARAMETER 2
CODE BS5950
TRACK 2 ALL
SBLT 1.0

PY 345000 ALL
CHECK CODE MEM 1401 TO 1424
LOAD LIST 2001 2002 2011 TO 2014
PRINT JOINT DISPLACEMENTS ALL
FINISH